suites, there are older cipher suites based on either RSA encryption or static Diffie-Hellman key agreement (or the elliptic curve variant, ECDH). Unfortunately, both algorithm families have significant security weaknesses, not directly related to their lack of forward secrecy. RSA key exchange uses an old mode of encryption (known as *PKCS#1 version 1.5*) that is very hard to implement securely and has resulted in many vulnerabilities in TLS implementations. Static ECDH key agreement has potential security weaknesses of its own, such as *invalid curve attacks* that can reveal the server's long-term private key; it is rarely implemented. For these reasons, you should prefer forward secret cipher suites whenever possible, because they provide better protection against common cryptographic vulnerabilities. TLS 1.3 has completely removed these older modes due to their insecurity.

> **DEFINITION** An ***invalid curve attack*** is an attack on elliptic curve cryptographic keys. An attacker sends the victim a public key on a different (but related) elliptic curve to the victim's private key. If the victim's TLS library doesn't validate the received public key carefully then the result may leak information about their private key. Ephemeral ECDH cipher suites (those with ECDHE in the name) are largely immune to invalid curve attacks because each private key is only used once.

Even if you use an older cipher suite, a DTLS implementation is required to include support for signatures in order to validate certificates that are presented by the server (and optionally by the client) during the handshake.